

Day 114 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in General Science (11): The GenomeIndia Project's Findings

READING MATERIAL

Covers the GenomeIndia Project's major findings — India's most comprehensive indigenous genomic study.

Q: When was the GenomeIndia Project launched, and by which body?

A: Launched in January 2020, led by the Department of Biotechnology under the Ministry of Science and Technology.

Q: How many people and population groups did the project's final sequencing cover?

A: 10,074 people, from 83 population groups — including tribal, rural, and urban communities, with 20+ leading research institutes (IISc Bengaluru, IITs, CSIR labs, genomics centers) participating.

Q: What did the project find in terms of genetic variants?

A: Nearly 130 million genetic variants total, including more than 44 million variants NEVER recorded in global genetic databases before.

Q: Why is this significant for medicine in India, specifically?

A: The project found 38 important genetic variants linked to how the body processes drugs (pharmacogenomic variants) — significant because most medicines used in India were developed based on genetic databases built primarily on European and North American genomes, not Indian populations.

Q: What infrastructure milestone has genomic sequencing reached in 2026?

A: The cost of Whole Genome Sequencing (WGS) dropped below \$200 — making it cheaper than (or comparable to) a traditional PET scan, with sequencing now available through a nationwide network of regional hubs, private diagnostics providers, and biobanks (not just elite academic labs).

MOCK TEST (WITH ANSWERS)

1. The GenomeIndia Project was launched in January 2020, led by which body?

A. ICMR

B. Department of Biotechnology ✓

C. DRDO

D. CSIR alone

Explanation: Department of Biotechnology, Ministry of Science and Technology. [medium, web-research]

2. The GenomeIndia Project's final sequencing covered how many people, from how many population groups?

A. 5,000 people, 50 groups

B. 10,074 people, 83 groups ✓

C. 20,000 people, 100 groups

D. 50,074 people, 150 groups

Explanation: 10,074 people, 83 population groups. [hard, web-research]

3. Of the nearly 130 million genetic variants found, how many had never been recorded in global genetic databases before?

A. 14 million

B. 24 million

C. 34 million

D. 44 million ✓

Explanation: 44 million. [hard, web-research]

4. Why is the project significant for Indian medicine specifically?

A. It found a cure for diabetes

B. Most existing drug-related genetic databases were built on European/North American genomes, not Indian populations ✓

C. It eliminated the need for clinical trials

D. It only studied animal genomes

Explanation: Existing pharmacogenomic databases weren't representative of Indian populations. [medium, web-research]

5. In 2026, the cost of Whole Genome Sequencing (WGS) dropped below:

A. \$100

B. \$200 ✓

C. \$500

D. \$1,000

Explanation: \$200. [medium, web-research]

Part B – Aptitude & Mental Ability: Percentage

READING MATERIAL

Percentage is 1 of 10 named Aptitude sub-skills. TNPSC's percentage questions are built from a handful of recurring shapes — successive discounts, reverse percentage ('20% of $x = 50$, find x '), percentage of a mixture, and percentage of a time period — recognize the shape first, then compute.

Q: Method — successive percentage discounts: A TV was sold for ₹14,400 after successive discounts of 10% and 20%. What was its original (market) price?

A: Successive discounts don't add (it's NOT a flat 30% off) — apply them one after another. If the market price is P : $P \times 0.90 \times 0.80 = 14,400 \rightarrow P \times 0.72 = 14,400 \rightarrow P = 14,400 / 0.72 = ₹20,000$.

Q: Method — percentage of a mixture (the parts must add to 100%): An alloy is 30% copper and 40% zinc, rest nickel. Find the nickel in 20 kg of the alloy.

A: Nickel % = $100 - 30 - 40 = 30\%$. Nickel amount = 30% of 20 kg = 6 kg. The key step is realizing the percentages must total 100% — nickel is whatever's left over, not a number given directly.

Q: Method — reverse percentage ('find the whole, given a percentage of it'): If 20% of $x = 50$, find x .

A: 20% of $x = 50 \rightarrow 0.20 \times x = 50 \rightarrow x = 50 / 0.20 = 250$. Reverse-percentage questions give you the part and the percentage, and ask for the whole — divide, don't multiply.

Q: Method — percentage of a decimal/small number: Find 0.75% of 40 kg.

A: $0.75\% = 0.0075$. $0.0075 \times 40 = 0.3$ kg. Common mistake: treating '0.75%' as '0.75' (i.e. 75%) instead of converting the decimal percentage properly — always convert to a decimal fraction first (divide by 100) before multiplying.

Q: Method — percentage of a time period: What percentage of a day is 10 hours?

A: A day = 24 hours. $10/24 \times 100 = 41.67\%$.

Q: Method — percentage of a fraction: What percent of $2/7$ is $1/35$?

A: Set up: $x\% \text{ of } (2/7) = 1/35 \rightarrow x/100 \times 2/7 = 1/35 \rightarrow x = (1/35) \times (7/2) \times 100 = (7/70) \times 100 = 10\%$.

Q: Method — sum of two independent percentage-of-number terms: Find the value of 28% of 450 + 45% of 280.

A: 28% of 450 = 126. 45% of 280 = 126. Sum = $126 + 126 = 252$. (A nice coincidence both terms equal 126 here — don't expect that in general, just add the two percentage values separately.)

Q: Method — percentage greater than 100% (the result is bigger than the original number): Find 280% of 3,940.

A: $280\% = 2.80$. $2.80 \times 3,940 = 11,032$. Percentages over 100% just mean the result is larger than the base number — the calculation method doesn't change.

MOCK TEST (WITH ANSWERS)

1. A mobile phone was sold for ₹10,200 after successive discounts of 15% and 20%. What was its original price?

- A. ₹14,500
- B. ₹15,000 ✓**
- C. ₹15,500
- D. ₹16,000

Explanation: $P \times 0.85 \times 0.80 = 10,200 \rightarrow P \times 0.68 = 10,200 \rightarrow P = 15,000$. [medium, authored]

2. An alloy is 25% tin and 35% lead, the rest is copper. Find the copper in 40 kg of the alloy.

- A. 10 kg
- B. 14 kg
- C. 16 kg ✓**
- D. 20 kg

Explanation: Copper % = $100 - 25 - 35 = 40\%$. 40% of 40 kg = 16 kg. [easy, authored]

3. If 35% of $y = 91$, find y .

- A. 230
- B. 250
- C. 260 ✓**

D. 240

Explanation: $y = 91 / 0.35 = 260$. [easy, authored]

4. Find 1.25% of 80 kg.

A. 1 kg ✓

B. 10 kg

C. 0.1 kg

D. 12.5 kg

Explanation: $1.25\% = 0.0125$. $0.0125 \times 80 = 1$ kg. [easy, authored]

5. What percentage of a day is 6 hours?

A. 20%

B. 25% ✓

C. 30%

D. 16.67%

Explanation: $6/24 \times 100 = 25\%$. [easy, authored]

6. What percent of $3/8$ is $3/40$?

A. 10%

B. 15%

C. 20% ✓

D. 12%

Explanation: $x\%$ of $(3/8) = 3/40$, so $x/100 = (3/40) \div (3/8) = (3/40) \times (8/3) = 1/5$, giving $x = 20\%$. [medium, authored]

7. Find the value of 32% of 350 + 18% of 200.

A. 148 ✓

B. 138

C. 150

D. 146

Explanation: 32% of 350 = 112. 18% of 200 = 36. Sum = 148. [medium, authored]

8. Find 320% of 2,500.

A. 7,500

B. 8,000 ✓

C. 8,500

D. 9,000

Explanation: $320\% = 3.2$. $3.2 \times 2,500 = 8,000$. [easy, authored]

9. A quantity is first increased by 20%, then decreased by 20%. What is the net percentage change?

A. No change

B. 4% decrease ✓

C. 4% increase

D. 20% decrease

Explanation: Net change after +20% then -20% is always a decrease of $(20 \times 20)/100 = 4\%$ — increasing and then decreasing by the same percentage never cancels out, because the second change applies to a different (larger) base. [hard, authored]

10. The price of an item increases from ₹250 to ₹300. What is the percentage increase?

A. 15%

B. 20% ✓

C. 25%

D. 30%

Explanation: Increase = 50. Percentage increase = $(50/250) \times 100 = 20\%$. [easy, authored]

Part C – General English: Prose: "Empowered Women Navigating the World"

READING MATERIAL

A real, recent (2017-18) non-fiction account of 'Navika Sagar Parikrama' — the Indian Navy's first all-women crew to circumnavigate the globe non-stop, aboard the indigenously-built sailboat INSV Tarini, showcasing women's empowerment and the 'Make in India' initiative.

Q: What does INSV stand for, what kind of vessel is INSV Tarini, and after whom is it named?

A: INSV = Indian Naval Ship Vessel. INSV Tarini is a 55-foot sailboat. It is named after Tara-Tarini, regarded as the patron deity for sailors.

Q: How many crew members undertook the voyage, how long did the voyage last, and what renewable energy source did the vessel use?

A: Six all-women crew members. The voyage lasted 254 days. The vessel used wind energy (it is a sailboat).

Q: Where and how long were the crew trained, and who mentored them?

A: They trained in Mumbai and Kochi, over a period of 3 years. They were mentored by Commander Dilip Donde (himself the first Indian to complete a solo non-stop circumnavigation).

Q: What was the name of the overall project/mission, and what national initiative did it aim to showcase?

A: The mission was called 'Navika Sagar Parikrama'. It aimed to showcase women's empowerment and the indigenous 'Make in India' initiative (since INSV Tarini was built in India).

Q: What quality of the skipper was credited with the expedition's success, and what value did one crew member (Payal Gupta) particularly highlight?

A: Honesty was credited as the skipper's key quality for successful leadership. Payal Gupta spoke about the importance of teamwork during the voyage.

Q: What festival did the crew celebrate while at sea, illustrating their resilience and team spirit far from home?

A: Diwali.

MOCK TEST (WITH ANSWERS)

1. INSV stands for:

- A. Indian National Ship Vessel
- B. Indian Naval Ship Vessel ✓**
- C. Indian Navy Service Vehicle
- D. Indian Nautical Sail Vessel

Explanation: Indian Naval Ship Vessel. [easy, web-research]

2. INSV Tarini is named after Tara-Tarini, who is regarded as the:

- A. Goddess of wisdom
- B. Patron deity for sailors ✓**
- C. Goddess of wealth
- D. Patron deity of warriors

Explanation: Patron deity for sailors. [medium, web-research]

3. How many members were in the all-women crew of INSV Tarini?

- A. Four
- B. Five
- C. Six ✓**
- D. Seven

Explanation: Six. [medium, web-research]

4. The crew underwent training for this voyage at:

- A. Chennai
- B. Kolkata
- C. Mumbai and Kochi ✓**
- D. Delhi and Pune

Explanation: Mumbai and Kochi. [medium, web-research]

5. The crew's training for the voyage lasted approximately:

- A. 6 months
- B. 1 year
- C. 2 years
- D. 3 years ✓**

Explanation: 3 years. [medium, web-research]

6. The crew was mentored by:

- A. Captain Rakesh
- B. Lt. Sinha
- C. Commander Dilip Donde ✓**
- D. Admiral Mehta

Explanation: Commander Dilip Donde. [medium, web-research]

7. The Indian Navy's all-women circumnavigation voyage aboard INSV Tarini lasted how many days?

- A. 180
- B. 200
- C. 254 ✓**
- D. 300

Explanation: 254 days. [easy, web-research]

8. What kind of vessel is INSV Tarini, and what renewable energy powers it?

- A. A cargo ship powered by solar energy
- B. A submarine powered by hydro energy
- C. A 55-foot sailboat powered by wind energy ✓**
- D. An oil tanker powered by biomass

Explanation: A 55-foot sailboat, powered by wind. [medium, web-research]

9. Which quality of the skipper is credited with the expedition's success?

- A. Bravery
- B. Strength
- C. Honesty ✓**
- D. Discipline

Explanation: Honesty. [medium, web-research]

10. The mission name for this all-women circumnavigation voyage was:

- A. Sagar Mitra
- B. Navika Sagar Parikrama ✓**
- C. Samudra Setu
- D. Nau Sena Abhiyan

Explanation: Navika Sagar Parikrama. [medium, web-research]